

# Sebastian Parakal Jose

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## Summary

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Results-driven software engineer and postgraduate student at Monash University (GPA 4.0/4.0) with 2+ years of industry experience in backend systems, cloud infrastructure, and mobile development. Delivered a globally launched platform serving **30,000+** users on day one and currently mentoring **100+** students as a Teaching Associate. Actively seeking full-time software engineering roles. **Unlimited work rights in Australia.**

## Education

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### Monash University

Melbourne, Australia

*Master of Information Technology*

*Jul 2024 – Jun 2026*

- **GPA: 4.0/4.0 (WAM: 86)**

- High Distinction in Cloud Computing and Security, Mobile and Distributed Computing Systems, Project Management, Advanced Database Technology, Internet Applications Development, Foundations of Data Science, and Software Engineering

### National Institute of Technology Calicut

Kerala, India

*Bachelor of Technology in Computer Science and Engineering (First Class)*

*2018 – 2022*

## Experience

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### Chief Technology Officer (CTO)

Jan 2026 – Present

*Monash Students for AI with Communities (MOSAIC)*

*Melbourne, Australia*

- Lead development of community-focused AI platforms and coordinate engineering teams across multiple student developers
- Directed the production launch of the **IFRC** World Disaster Report platform ([wdr26.org](http://wdr26.org)), launched globally in Geneva on March 5 supporting over **30,000** website users on day one
- Introduced AI-assisted development workflows using **Claude**, **GitHub Copilot**, and **Cursor**, and designed developer onboarding workflows in **Notion**, standardizing practices and improving team ramp-up efficiency and delivery speed

### Teaching Associate

Jul 2025 – Present

*Monash University*

*Melbourne, Australia*

- Support teaching for FIT5046 Mobile and Distributed Computing Systems and FIT1056 Introduction to Software Engineering
- Mentor **100+** students in Android development using Jetpack Compose, MVVM architecture, and REST API integration
- Designed and built a multi-agent **Claude Code** marking harness for Assessment 4 group projects, orchestrating five specialized sub-agents (code-inspector, report-auditor, rubric-scorer, uniqueness-checker, verifier) for context-grounded, evidence-bound rubric scoring with independent mismatch verification (hallucination/wrong-file/arithmetic/scope-creep classification), cutting manual cross-checking time while preserving a full audit log
- Achieved a **91%** student satisfaction rating (SETU) through structured mentoring and technical support

### Software Engineer

Jul 2022 – Jul 2024

*NCR Corporation / Global Payments*

*Hyderabad, India*

- Engineered and maintained mission-critical **C++** backend services within the payment processing platform, supporting global financial networks including **Mastercard** and **American Express** across ATM and payment switch systems
- Implemented and enhanced **ISO8583** transaction messaging pipelines handling high-volume financial message routing between card networks, acquirers, and issuing banks
- Contributed to the **L4 escalation team**, diagnosing and resolving the most complex production incidents escalated from lower support tiers; identified root causes in payment processing logic and deployed hotfixes under live system constraints, improving overall system reliability by **20%**
- Used **Splunk** to build monitoring dashboards and queries for real-time pipeline observability, reducing mean time to detect (MTTD) production anomalies by **30%** and accelerating root cause analysis
- Optimized **Oracle SQL** queries and stored procedures underpinning transaction records, improving processing latency by **25%** across peak-load periods

## Projects

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### Hush | [hushapp.live](https://hushapp.live)

*Vue.js, AWS Amplify, Lambda, RDS PostgreSQL, ECR, Terraform, EventBridge, Playwright, Vitest, GitHub Actions*

- Built a production city navigation platform for neurodivergent users at **hushapp.live**, addressing sensory overload in urban environments by providing real-time crowd density mapping and quiet route planning across Melbourne's CBD
- Architected a serverless backend with AWS Lambda, API Gateway, and **ECR**-hosted container images; provisioned all infrastructure as code using **Terraform** with least-privilege **IAM** roles per Lambda function, and scheduled automated data refresh via **EventBridge** with strict **CORS** enforcement
- Established a full CI/CD pipeline using **GitHub Actions** running **Vitest** unit tests and **Playwright** end-to-end tests on every pull request, with **Dependabot** automating dependency security updates

### SignalStack | *Python, Kafka (KRaft), PostgreSQL, Redis, FastAPI, Next.js, Terraform, AWS, Prometheus, Grafana*

- Architected **SignalStack**, a self-hosted tech-trend platform ingesting real-time activity from **GitHub** and **Hacker News**, scoring trending CS topics via a custom **Apache Kafka (KRaft)** pipeline on a single free-tier AWS EC2 instance
- Built a fault-tolerant consumer pipeline with partitioned Kafka topics, batched **PostgreSQL** writes, and **Redis** deduplication, raising throughput ~250x at zero added infrastructure cost
- Served data via a 9-endpoint **FastAPI** service with Redis caching and a **Next.js** frontend and provisioned with **Terraform**, automated CI/CD via **GitHub Actions**, and monitored with self-hosted **Prometheus/Grafana**

### Pose Estimation Web Service | *FastAPI, Docker, Kubernetes, YOLOv8, Locust*

- Built a cloud-native pose detection API using **YOLOv8** to enable real-time human movement analysis for fitness and physiotherapy applications, designed to operate under strict resource constraints (512MiB memory, 0.5 CPU)
- Optimized the inference pipeline through image resizing, garbage collection, and concurrency control, reducing memory usage by **35%** and improving throughput **2x** across horizontally scaled Kubernetes pods
- Containerized with **Docker** and deployed on **Kubernetes** with horizontal pod autoscaling; validated performance under load using **Locust** to simulate concurrent real-world user traffic

### BirdTag | *AWS Lambda, S3, DynamoDB, Cognito, API Gateway, ECR*

- Built a serverless media management platform for wildlife researchers and citizen scientists to tag, search, and retrieve bird observation recordings, replacing manual file management with structured metadata-driven queries
- Integrated **S3** for media storage, **Cognito** for user authentication, and **API Gateway** with **Lambda** for metadata operations; designed **DynamoDB** schema for low-latency retrieval across large observation datasets
- Packaged AI species-identification models as **ECR** container images consumed by Lambda, enabling scalable serverless inference without managing dedicated model-serving infrastructure

### PantryPals | *Kotlin, Jetpack Compose, Room, Retrofit, WorkManager, AlarmManager, ML Kit*

- Developed an Android app to combat household food waste (estimated at \$2,500/year per Australian family) by tracking product expiry dates through **ML Kit** barcode scanning and OCR-based expiry detection
- Built offline-first storage using **Room** database with **Retrofit**/OpenFoodFacts API integration for automatic product data retrieval, and analytics dashboards visualising pantry usage trends
- Implemented background expiry monitoring via **WorkManager** and precise user notifications via **AlarmManager**; integrated fridge sensor data to surface real-time temperature alerts alongside expiry warnings

## Technical Skills

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**Languages:** Python, Java, Kotlin, C++, SQL, R

**Backend & Frameworks:** FastAPI, Flask, Node.js, REST APIs, Apache Kafka, Jetpack Compose, Retrofit

**Cloud & Infrastructure:** AWS (Lambda, API Gateway, S3, DynamoDB, Cognito, ECR, RDS, EC2, Amplify, EventBridge), Docker, Kubernetes, Terraform

**Databases & Data Systems:** PostgreSQL, Oracle SQL, DynamoDB, Firebase, Room Database, Redis, PostGIS

**DevOps & CI/CD:** Git, GitHub Actions, Dependabot, Jenkins, SonarQube

**Testing:** Playwright (E2E), Vitest (Unit), Locust (Load), Regression Testing

**Security:** CORS, HTTPS/HSTS, XSS Prevention, Role-Based Access Control

**Observability & Monitoring:** Splunk, Prometheus, Grafana

**Frontend & Web:** VueJS, Next.js, Vite, HTML, CSS

**Machine Learning & AI:** YOLOv8, Computer Vision

**AI-Assisted Development:** Claude, Claude Code, GitHub Copilot, Cursor, Multi-Agent Orchestration, Prompt Engineering, Harness Engineering